

The Hodgkin–Huxley membrane, in equations

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Almost all equations, but not quite: a few lines of prose between the blocks so the eye has somewhere to rest. The subject is the four-dimensional system for a space-clamped patch of excitable membrane, the model Hodgkin and Huxley fit to the squid giant axon in 1952 [1], with state $\mathbf{x} = (V, m, h, n)^\top$: one voltage and three gating variables. Everything below is that one system, taken apart one relation at a time.

Current balance

The membrane is a capacitor in parallel with three voltage-gated conductances [1, 10]. Charge conservation across it gives the equation that ties the whole model together: the rate of change of voltage is set by the sum of ionic currents plus whatever is injected:

$$C_m \dot{V} = - \underbrace{\bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}})}_{I_{\text{Na}}} - \underbrace{\bar{g}_{\text{K}} n^4 (V - E_{\text{K}})}_{I_{\text{K}}} - \underbrace{g_{\text{L}} (V - E_{\text{L}})}_{I_{\text{L}}} + I_{\text{ext}}(t)$$

C_m , membrane capacitance; $\bar{g}_\bullet$, peak conductances; $E_\bullet$, reversal potentials; $m, h, n \in [0, 1]$, gating variables; I_{ext} , injected current.

Each conductance is the peak value scaled by gates raised to a power: the exponents (m^3 , n^4) are the number of independent subunits that must all be open at once. The gates are what make the patch excitable; the next few blocks say how they move.

Gating kinetics

Each gate relaxes toward a voltage-dependent target, written two equivalent ways, as forward/backward rates, or as a target value x_∞ approached with time constant τ_x :

$$\dot{x} = \alpha_x(V)(1 - x) - \beta_x(V)x = \frac{x_\infty(V) - x}{\tau_x}(V), \quad x_\infty = \frac{\alpha_x}{\alpha_x + \beta_x}, \quad \tau_x = \frac{1}{\alpha_x + \beta_x}, \quad x \in \{m, h, n\}.$$

$\alpha_x(V)$, $\beta_x(V)$, voltage-dependent opening/closing rate constants; x_∞ , steady-state (in)activation; τ_x , its relaxation time constant; $x \in \{m, h, n\}$, the three gating variables.

Rate functions (V in mV, rates in ms^{-1})

The rates themselves are the empirical heart of the model: six functions fit to voltage-clamp data, no first-principles derivation behind them [1]. They are what turns the abstract kinetics above into concrete dynamics:

$$\begin{aligned} \alpha_m(V) &= \frac{0.1(V + 40)}{1 - \exp(-(V + 40)/10)}, & \beta_m(V) &= 4 \exp(-(V + 65)/18), \\ \alpha_h(V) &= 0.07 \exp(-(V + 65)/20), & \beta_h(V) &= \frac{1}{1 + \exp(-(V + 35)/10)}, \\ \alpha_n(V) &= \frac{0.01(V + 55)}{1 - \exp(-(V + 55)/10)}, & \beta_n(V) &= 0.125 \exp(-(V + 65)/80). \end{aligned}$$

The α_m and α_n expressions look singular: numerator and denominator both vanish at one voltage, so a naive evaluation returns 0/0. The function is smooth there anyway: the singularity is removable, and the value is just the limit [9]:

$$\alpha_m(V) = \begin{cases} \frac{0.1(V+40)}{1 - \exp(-(V+40)/10)} & \text{if } V \neq -40 \\ 1 & \text{if } V = -40 \end{cases} \quad \text{since} \quad \lim_{V \rightarrow -40} \alpha_m(V) = 1.$$

Steady-state gating as a Boltzmann sigmoid

Plotted against voltage, each x_∞ is an S-curve, and it is often more convenient to fit it directly as a Boltzmann sigmoid with a half-activation voltage $V_{1/2}$ and a slope factor k than to go through the rate functions [7, 8]. Its derivative has the tidy logistic form that makes the slope at $V_{1/2}$ exactly $1/(4k)$:

$$x_\infty(V) = \frac{1}{1 + \exp(-(V - V_{1/2})/k)}, \quad \frac{dx_\infty}{dV} = \frac{x_\infty(1 - x_\infty)}{k}.$$

$V_{1/2}$, half-activation voltage; k , slope factor.

Equilibrium potentials

The reversal potentials $E_\bullet$ that appear in the current balance are not free parameters: they are set by the ionic concentrations either side of the membrane. Nernst gives the single-ion equilibrium; Goldman–Hodgkin–Katz generalises to the current carried by a partially permeant ion and to the resting potential of a membrane permeable to several ions at once [2, 3]:

$$E_S = \frac{RT}{z_S F} \ln \frac{[S]_{\text{out}}}{[S]_{\text{in}}}, \quad I_S = P_S z_S^2 \frac{VF^2}{RT} \cdot \frac{[S]_{\text{in}} - [S]_{\text{out}} e^{-z_S VF/RT}}{1 - e^{-z_S VF/RT}},$$

$$V_m = \frac{RT}{F} \ln \frac{P_K[K]_{\text{out}} + P_{\text{Na}}[\text{Na}]_{\text{out}} + P_{\text{Cl}}[\text{Cl}]_{\text{in}}}{P_K[K]_{\text{in}} + P_{\text{Na}}[\text{Na}]_{\text{in}} + P_{\text{Cl}}[\text{Cl}]_{\text{out}}}.$$

E_S , Nernst potential of ion S ; V_m , resting potential of the mixed-permeability membrane; I_S , GHK current carried by S ; R , gas constant; T , absolute temperature; F , Faraday constant; z_S , valence of S ; $P_\bullet$, membrane permeabilities; $[S]_{\text{in}}$, $[S]_{\text{out}}$, intra- and extracellular concentrations.

Linear stability

With the vector field fully specified, the question of whether the rest state is quiet or poised to spike becomes a matter of eigenvalues [4, 5, 6]. Write the system as $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ and linearise about a fixed point $\mathbf{x}^*$ where $\mathbf{F}(\mathbf{x}^*) = \mathbf{0}$. The Jacobian $\mathbf{J} = \partial \mathbf{F} / \partial \mathbf{x}$ is

$$\mathbf{J} = \begin{bmatrix} \partial_V \dot{V} & \partial_m \dot{V} & \partial_h \dot{V} & \partial_n \dot{V} \\ \partial_V \dot{m} & -1/\tau_m & 0 & 0 \\ \partial_V \dot{h} & 0 & -1/\tau_h & 0 \\ \partial_V \dot{n} & 0 & 0 & -1/\tau_n \end{bmatrix}, \quad \partial_V \dot{V} = -\frac{g_L + \bar{g}_{\text{Na}} m^3 h + \bar{g}_{\text{K}} n^4}{C_m},$$

and the rest state is stable iff every eigenvalue has negative real part: $\det(\mathbf{J} - \lambda \mathbf{I}) = 0 \implies \text{Re}(\lambda_i) < 0$ for all i .

$\mathbf{F}$, the model's vector field; $\mathbf{x}^*$, a fixed point; $\mathbf{J}$, the Jacobian there; λ_i , its eigenvalues; $\partial_a \dot{b}$, the partial of $\dot{b}$ with respect to a .

Synaptic drive and subthreshold response

So far I_{ext} has been an anonymous forcing term. In a network it is synaptic: presynaptic spikes open conductances that pull the membrane toward a synaptic reversal potential [7]. A spike train $\{t_k\}$ arriving through a synapse of reversal E_{syn} adds

$$I_{\text{syn}}(t) = g_{\text{syn}}(t)(V - E_{\text{syn}}), \quad g_{\text{syn}}(t) = \bar{g} \sum_{k: t_k \leq t} \frac{t - t_k}{\tau_s} e^{1-(t-t_k)/\tau_s},$$

where each presynaptic spike contributes an alpha-function conductance transient of peak $\bar{g}$ (the synaptic weight) that rises and decays on the synaptic time constant τ_s . Below threshold, where

the membrane is a leaky integrator $\tau \dot{V} = -(V - V_{\text{rest}}) + RI$ with membrane time constant τ and input resistance R , the voltage is the convolution

$$V(t) = V_{\text{rest}} + \frac{R}{\tau} \int_{-\infty}^t e^{-(t-s)/\tau} I(s) ds, \quad \hat{Z}(\omega) = \frac{R}{1 + i\omega\tau},$$

with $\hat{Z}(\omega)$ the input impedance, a first-order low-pass whose cutoff sits at $\omega_c = 1/\tau$.

g_{syn} , synaptic conductance; E_{syn} , synaptic reversal potential; $\{t_k\}$, presynaptic spike times; $\bar{g}$, peak conductance (synaptic weight); τ_s , synaptic time constant; τ , R , membrane time constant and input resistance; $\hat{Z}(\omega)$, input impedance; ω_c , low-pass cutoff.

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